

37495 Statistical Design and Models for Evaluation Studies

Computer Lab 2

Question 1

Suppose that you want to compare the growth of tomatoes under different growing conditions. Identify relevant factors and any blocking factors that should be considered. Write a paragraph on the experiment design, identifying the treatments, experimental units, measurement units and an appropriate response variable.

Question 2

An experiment was conducted to test the effects of nitrogen fertiliser on lettuce production. Five rates of ammonium nitrate were applied to four replicate plots in a CRD and the number of heads of lettuce per plot counted.

```
dataQ2 <- read.table("dataQ2.txt", header=TRUE, colClasses=c("factor","numeric"))
```

Table 1: Heads of lettuce per plot

treatment	1	2	3	4
0	104	114	90	140
50	134	100	144	174
100	146	142	172	156
150	147	160	160	163
200	131	148	154	163

- Plot the data and comment on any relationships that you see.
- Write down the linear model for this study, including indices. Explain the model components.
- Compute the analysis of variance table for the data.
- What hypothesis is the test in the analysis of variance table testing?
- What are your conclusions from this analysis?
- Calculate $\hat{\tau}_i$ for $i \in \{1, \dots, 5\}$.
- This experiment was conducted in a completely randomised design with field plots in a 4×5 array of plots. Show a randomisation of the five treatments to the 20 plots. Explain how you obtained this randomisation.

Question 3

Four cultivars, A, B, C, and D, of dwarf beans were arranged in a CRD with eight replicates each. The layout of the design was as indicated below. The leaf areas (cm^2) were recorded, as shown, three days after germination. Three of the plots suffered damage from a windborne insecticide spray from a neighbouring field and these plants are indicated as missing (NA).

```
dataQ3 <- read.csv("dataQ3.csv", header=TRUE, colClasses=c("factor","numeric"))
```

To remove rows with “NA” values use the code below.

```
dataQ3 <- dataQ3[!is.na(dataQ3$area),]
```

Perform an appropriate analysis and comment on your findings.

Question 4

The table below gives the coagulation times for samples of blood drawn from 24 animals receiving four different diets, A, B, C, and D. The animals were randomly assigned to the four diets and samples were tested in a random order.

```
dataQ4 <- readRDS(file="dataQ4.RDS")
```

Table 2: Coagulation times

diet	1	2	3	4	5	6
A	62	60	63	59	63	59
B	63	67	71	64	58	66
C	68	66	71	67	68	68
D	56	62	60	72	63	64

- Plot the data and comment on any relationships that you see.
- Write down the linear model for this study, including indices. Explain the model components.
- Compute the analysis of variance table for the data.
- What hypothesis is the test in the analysis of variance table testing?
- What are your conclusions from this analysis?
- Why might we need to test the samples in a random order?